Theory of Computation (CS:4330:0001)
Homework 1

This assignment in based on the material covered in the first two weeks from Chapter 1 (Regular Languages). Before you start on this home work assignment, be sure to review the course policies on academic integrity and late submissions from the syllabus!

1. Consider a DFA M whose formal description is $(\{q_1, q_2, q_3, q_4, q_5\}, \{\mathsf{u}, \mathsf{d}\}, \delta, q_3, \{q_3\})$, where δ is given by the following table. Give the state diagram of this machine. Note that a DFA is simply what we have been calling a Finite Automaton (FA). (10 points)

	u	d
q_1	q_1	q_2
q_2	q_4	q_3
q_3	q_5	q_4
q_4	q_3	q_5
q_5	q_4	q_3

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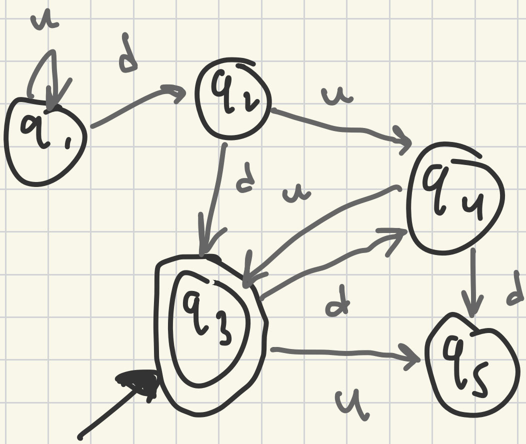
Late by 2 days due to sudden death in family; was absent tuesday - saturday week of 9/9/25 and could not complete until today.

$M = (Q, \Sigma, \delta, q_0, F)$

$(\{q_1, q_2, q_3, q_4, q_5\}, \{u, d\}, \delta, q_3, \{q_3\})$



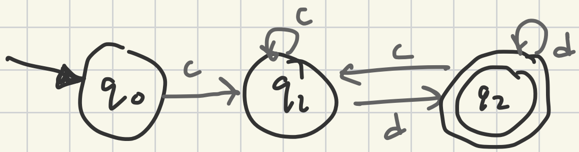
2. Give state diagrams of DFAs recognizing the following languages. In all parts, the alphabet is $\{\mathsf{c}, \mathsf{d}\}$. (12 points)

   (a) $\{w \mid w$ begins with a c and ends with a $\mathsf{d}\}$.
   (b) $\{w \mid w$ contains at least three c's$\}$.
   (c) $\{w \mid$ every odd position in w is a $\mathsf{d}\}$.
   (d) $\{w \mid$ the length of w is at most $4\}$.

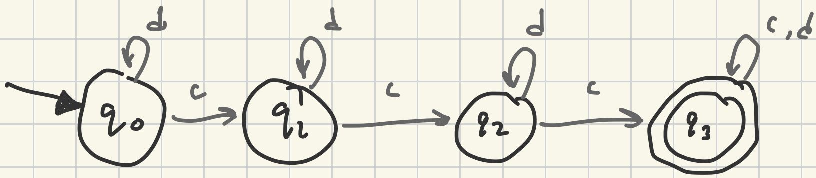
using q_n $n = 1, 2, 3, \ldots$

$q_n \in Q$

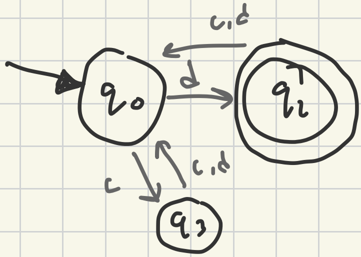
a) string begins with 'c' & ends with 'd'



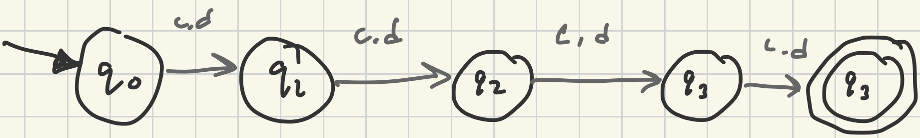
b) string contains at least 3 'c'



c) every odd symbol is a 'd'



d) $|w| \leq 4$ symbols



3. Let $B_n = \{c^k \mid k \text{ is a multiple of } n\}$. Show that for each $n \geq 1$, B_n is regular. Note that c^k denotes the string with k c's. You can also assume the alphabet has only the symbol c. **Hint**: First show that B_2 and B_3 are regular. Then generalize to any $n \geq 1$. (8 points)

$\mathbb{N}$ natural #s

$\mathbb{N} = \{1, 2, \ldots$

$B_n = \{c^k \mid k$ is a multiple of $n\}$

$\cancel{(\Sigma^* c \Sigma^*)}^k$: alphabet only has "c" $\Rightarrow c^k$

Proof: Induction

1. Base cases:

$n=1$: k is divisible by 1 ; $\forall n \in \mathbb{N} \quad 1 \mid n$

knowing $n \mid k$, transitive relationship :

$(1, n)\ (n, k)$ shows $1, k \Rightarrow 1 \mid k$

B_1 is regular □

$n=2$: k is divisible by 2 ;

$c^2 = (cc)^* \rightarrow |(cc)^*| = 2^z \qquad 2 \mid 2^z \therefore B_2$ regular □

$z \in \mathbb{Z}$

$n=3$: k is divisible by 3 ;

$c^3 = (ccc)^* \rightarrow |(ccc)^*| = 3^z \qquad 3 \mid 3^z \therefore B_3$ regular □

$z \in \mathbb{Z}$

2. Induction Step

$B_{n+1} = \{c^k \mid k$ is a multiple of $n+1\}$

Because we have shown $n \mid k$, we know that $k = zn$ thus

implicit Inductive Assumption

$k = z(n+1)$

$c^{z(n+1)} = (\underbrace{c \cdot c \cdot c \cdot c \ldots \cdot c}_{z(n+1)})^* = z(n+1)^q$

$q \in \mathbb{Z}$

thus, $(n+1) \mid z(n+1)$

$z(n+1) \mid z(n+1)^q \Rightarrow (n+1) \mid z(n+1)^q \quad \therefore B_n$ is regular □

4. Each of the following languages is the complement of a simpler language. In each part, construct a DFA for the simpler language, then use it to give the state diagram of a DFA for the language given. In all parts, the alphabet is $\{\mathsf{c}, \mathsf{d}\}$. (12 points)

(a) $\{w \mid w$ is any string except cd and $\mathsf{ddc}\}$.

(b) $\{w \mid w$ is any string that doesn't contain exactly two c's$\}$.

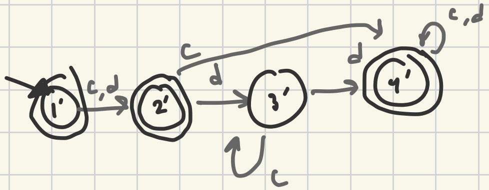
(c) $\{w \mid w$ is not a string of the form $\mathsf{d}^i\mathsf{c}^j$ for some $i, j \geq 0\}$.

* ↑ Assuming ε (empty string) included *

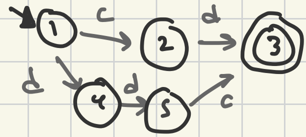
compliment machine: 1', 2', 3', 4' ...

machines: 1, 2, 3, 4 ...

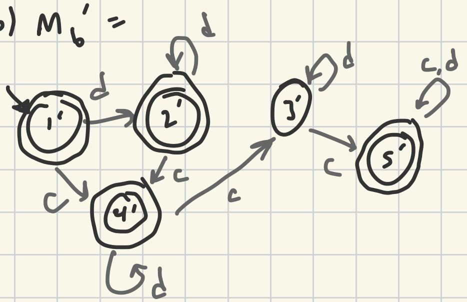
a) $M_a' =$



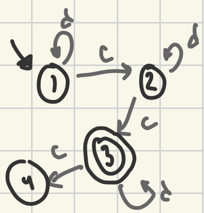
$M_a =$



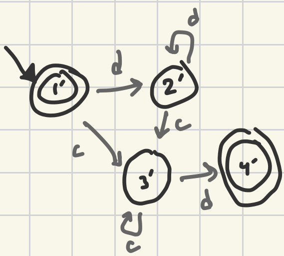
b) $M_b' =$



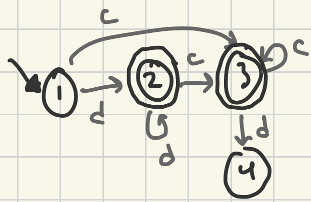
$M_b =$



c) $M_c' =$



M_c :



please read assumption...

This is for accepting state on starting state

5. The following L language is the intersection of two simpler languages. Construct DFAs for each of the simpler languages, then combine them using the construction discussed in Footnote 3 (page 46) to give the state diagram of the DFA for the language L. The alphabet is $\{c, d\}$. (8 points)

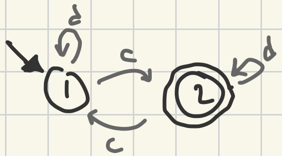
described by two languages

$$L = \{w \mid w \text{ has an } \underbrace{\text{odd number of c's}}_{L_1} \underbrace{\text{and}}_{\cap} \underbrace{\text{ends with}}_{L_2} \text{ a d}\}.$$

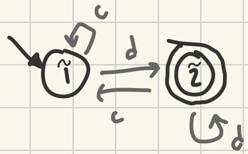
Footnote 3: "This expression would... M would accept the string only if both M_1 and M_2 accept it ... intersection and not union ... closed under intersection"

closure under intersection for $L = L_1 \cap L_2$:

$L_1 = \{w \mid w$ has odd number of c's$\}$



$L_2 = \{w \mid w$ ends with a d$\}$



construction of L:

$\Sigma = \Sigma_1 \cup \Sigma_2 = \{c, d\}$

$Q = \{(r_1, r_2) \mid r_1 \in Q_1 \text{ and } r_2 \in Q_2\} = Q_1 \times Q_2 = \{(1,\tilde{1}), (2,\tilde{1}), (1,\tilde{2}), (2,\tilde{2})\}$

$q_0 = (q_1, q_2) = (1, \tilde{1})$

$F = (F_1 \times F_2) = (2, \tilde{2})$

$\delta: \delta((r_1, r_2), a) = (\delta_1(r_1, a), \delta_2(r_2, a))$

(q_1, q_2)	c	d
$(1, \tilde{1})$	$(2, \tilde{1})$	$(1, \tilde{2})$
$(2, \tilde{1})$	$(1, \tilde{1})$	$(2, \tilde{2})$
$(1, \tilde{2})$	$(2, \tilde{1})$	$(1, \tilde{2})$
$(2, \tilde{2})$	$(1, \tilde{1})$	$(2, \tilde{2})$ *

DFA Diagram on next page ⇒

M such that M accepts $L = L_1 \cap L_2$